

C.4 Similarity analysis

In the examples above, we have shown that MACE-MP-0 is capable of surprising degrees of extrapolation. The use of semi-local features (as a result of message passing) (23) and element mixing (24) within the MACE architecture are key components underlying MACE-MP-0’s capabilities. These components allow MACE-MP-0 to extrapolate to systems that globally seem completely different from the training data but have close matches locally.

To quantify the similarity, we compare atomic environments from test systems to filtered portions of the training data using the following procedure:

1. Filter training data to a subset with elemental compositions similar or exactly matching the test system.
2. Use MACE-MP-0 to extract invariant descriptors for all atoms in both the test system and the filtered training subset.
3. Calculate the cosine similarity between the atoms in the test system and each filtered training structure. For each atom, we use the maximum cosine similarity found this way. This is essentially a best-match structure kernel (328) that allows many-to-one mappings.
4. Average these maximum atomic similarities by element and then combine them by averaging again, yielding an element-stratified similarity.

Through this procedure, we identify training set structures that contain the most similar local environments to those in any given test system. In addition, we create `chemiscope` (329) (<https://chemiscope.org/>) input files containing UMAP (330) projections of the atomic descriptors (fitted only on the training environments), allowing a more granular and interactive inspection of the environments in the test and training data.

The code for analysing the data and generating `chemiscope` inputs is available as a Python package (43).

D Uncertainty quantification

D.1 Theory and implementation

Predictive uncertainties for MACE-MP-0b3 are obtained with the last-layer prediction rigidity (LLPR) method (331). This approach is well-suited for MACE-MP-0b3 as it is simple, scalable, and allows to obtain uncertainty for neural networks that have already been trained. In the simplest form, LLPR uncertainties are given by

$$\sigma_i^2 = \alpha^2 \mathbf{f}_i^\top \mathbf{H}_o^{-1} \mathbf{f}_i, \quad (\text{S13})$$

where $\mathbf{f}_i$ is the hidden features of structure i in the last layer of the neural network (before entering the final linear transformation), $\mathbf{H}_o$ is the generalized Gauss-Newton pseudo-Hessian of the loss at the end of training, α^2 is a calibration parameter, and σ_i^2 is the resulting LLPR uncertainty for structure i .

In calculating the $\mathbf{H}_o$ for MACE-MP-0b3, only the energy part of the loss function is taken into account, while the force and stress terms are omitted. This greatly simplifies the practical calculation of the pseudo-Hessian, and—as we show in Fig. S72—still allows one to obtain high-quality error estimates on forces and stresses. In simple cases, LLPR uncertainties can easily be analytically propagated to the derived targets. A simple example is given by the subtraction of two different energies, which is later used to generate uncertainties for several NEB experiments in Section D.3. In such cases, the uncertainty in the difference between the energies of structures i and j is computed as $\alpha^2(\mathbf{f}_i - \mathbf{f}_j)^\top \mathbf{H}_o^{-1}(\mathbf{f}_i - \mathbf{f}_j)$. Other cases can be treated similarly by making use of the connection (331) between the LLPR method and the Laplace approximation (332, 333).

In terms of computational cost, the energy uncertainties obtained with (S13) generate negligible overhead when compared to simple energy predictions of MACE architecture. Predicting the error on gradients (e.g. forces, stresses) is more demanding, as it requires performing multiple backpropagation steps. However, this is only needed when explicit estimates of the gradient error, or of properties that depends explicitly on the gradient values, are desired. A LLPR-wrapped version of MACE-MP-0b3 that can output uncertainties for energy, force, and stress predictions are made available alongside the original model.

D.2 Distribution of errors in the validation set

Figure S72 shows the distributions of the predicted and actual errors on the forces and stresses of the validation set used in the training of MACE-MP-0b3 potential. Energies are omitted as the limited size of the validation set (160 structures) is not sufficient to gather significant statistics. Nonetheless, it is clear that the LLPR method recovers good-quality uncertainty predictions in both cases, despite the exclusion of forces and stresses terms from the pseudo-Hessian calculations. See Section D.3.1 for results on out-of-domain energy predictions.

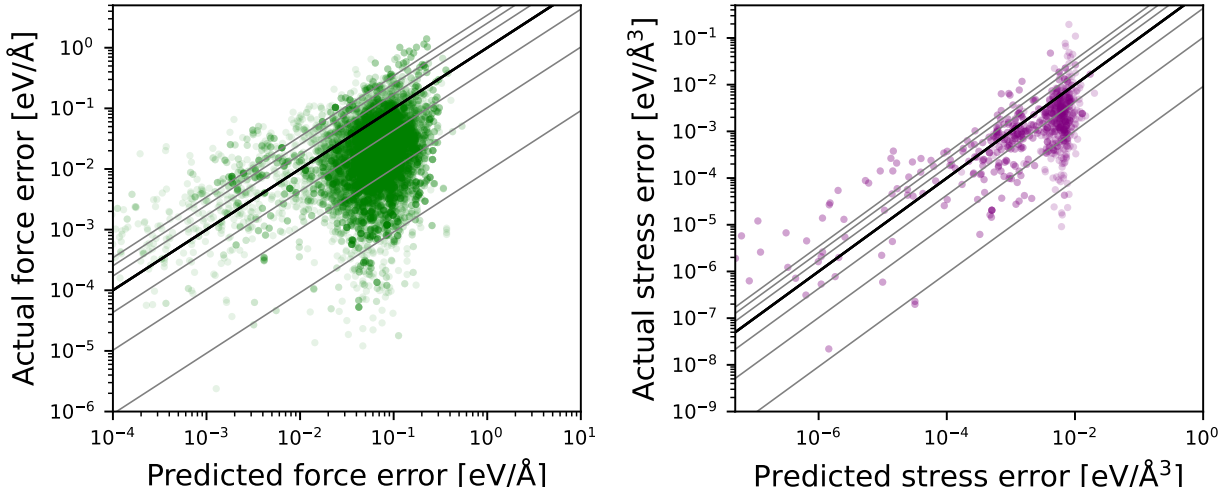


Figure S72: Force (left) and stress (right) errors on the validation structures for MACE-MP-0b3. The black line represents $y = x$, while the grey lines are pairs of isolines enclosing approximately 68%, 95% and 99% of the expected distribution, respectively. Errors on very small predictions (which are present due to the symmetry of several validation structures) are cut out, but they are nonetheless well-predicted by the LLPR approach. See Section D.3.1 for more details on these plots and their expected distribution.

D.3 Case studies

In this section, we show the LLPR uncertainties for a subset of the case studies found in section A. When interpreting these results, it is important to keep in mind that only the WBM dataset (section D.3.1) provides reference DFT energies that are consistent with the dataset MACE-MP-0b3 has been trained on. This is not the case for the other case studies considered, which makes it difficult to make any quantitative assessments, and in particular to pinpoint whether the discrepancy between the DFT reference data and the MACE-MP-0b3 predictions stems from the error in model prediction or the inconsistencies in the DFT methods employed. Nonetheless, the following subsections demonstrate that performing uncertainty quantification can still be insightful in detecting where the model succeeds or fails, even in the absence of consistent DFT calculations. Note that uncertainties for the NEB case studies are demonstrated using the NEB frames acquired with a previous generation of the model (MACE-MP-0a).

D.3.1 Uncertainty quantification for the WBM dataset

As discussed in section A.29, the WBM dataset (278) can be regarded as an out-of-distribution test set for MACE-MP-0b3. The WBM dataset also provides reference DFT energies that are consistent with that of the MPtrj dataset which the MACE-MP-0b3 has been trained on. We hence demonstrate the efficacy of the LLPR-based error estimates by performing uncertainty quantification for the total energies of the structures in the WBM dataset. Figure S73 shows the distribution of the absolute errors versus the estimated errors.

It should be understood that the estimated error corresponds to the width of the distribution of the prediction errors, and that it does not directly correlate with the actual absolute error. This is apparent

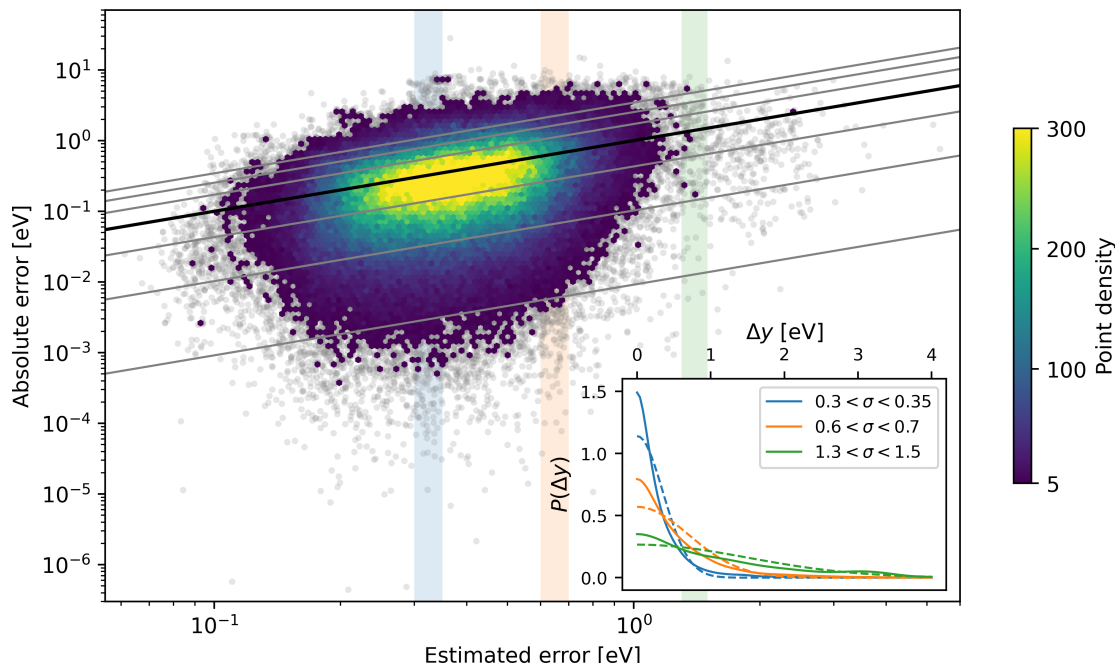


Figure S73: Scatter plot that shows the absolute error vs. estimated error in the total energies for the WBM dataset of Wang et al. (278) $y = x$ is shown in black, and isolines that successively bound 68%, 95%, and 99% of the distribution are shown in gray. Scatter plot for the entire dataset (light gray) is shown together with its 2D histogram. The inset shows the probability densities of points within three “slices” (highlighted in the main panel) of estimated errors. Solid line shows the probability density within a given slice estimated by Gaussian kernel density estimation. Dashed line shows the actual Gaussian distribution using the mean estimated error of the slice.

from the probability densities of the vertical “slices” (the inset of Figure S73) in the distribution, where the observed probability densities closely match that of the expected Gaussian distributions with σ values corresponding to each slice. One can see that a majority of the empirical errors lie within 3σ of the estimated error, and there is a limited number of statistically significant underestimations.

D.3.2 Uncertainty quantification for elemental defects in alumina

In section A.14.2 (Figure S24d-e), it is shown that MACE-MP-0b3 accurately recovers the reference NEB barrier path of an elemental defect diffusion for Y, whilst failing to do so for Co. In Figure S74, we show that LLPR-based uncertainty quantification can correctly discern the model accuracy between these two cases. The uncertainties for the case of Y remain small in the beginning and end of the NEB path, and only shows notable uncertainty in the middle of the path. On the contrary, the estimated uncertainties for the Co NEB barrier path is considerably large throughout the entire NEB path, even close to the beginning and end. In fact, it is generally larger by a factor of 4 when compared with that of Y.

D.3.3 Uncertainty quantification for CO oxidation on Cu

Section 2.2 and section A.24 demonstrates the application of MACE-MP-0b3 for CO oxidation on several different surfaces of Cu. Figure S75 shows the results for the first half of the pathway accompanied by the LLPR-based uncertainties. In all cases, LLPR uncertainties generally remain small for the reaction pathway up to coupling at (i). Despite inconsistencies in the DFT details, this is indeed where the MACE-MP-0b3 predictions remain close to the available DFT data. Past this range in the reaction coordinate, the uncertainties become larger in all cases, and the actual errors with respect to the available DFT data also become more severe.

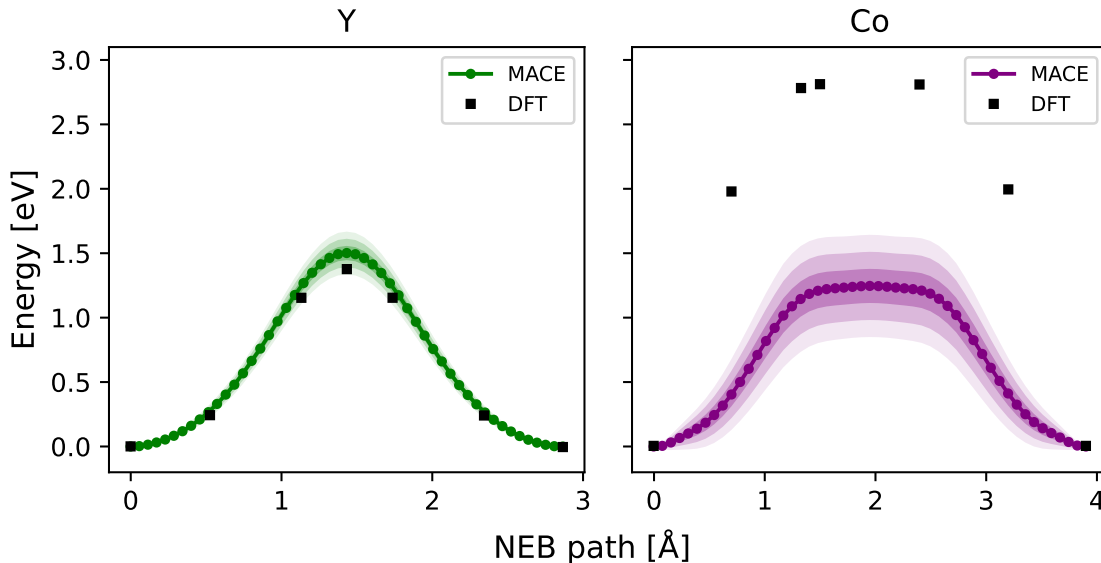


Figure S74: MACE-MP-0b3-predicted relative energies along the NEB barrier paths of elemental defect diffusion in alumina, for Y (left) and Co (right). Estimated error bounds are shaded along the plot, where 1σ , 2σ , and 3σ bounds are shown in successively lighter shades. Available DFT data (that is not entirely consistent with the MPtrj dataset) is shown with black square markers.

D.3.4 Uncertainty quantification for carborane rearrangement

Section A.25 discusses the capability of MACE-MP-0b3 to qualitatively capture the energetics of carborane rearrangement. Figure S76 shows that LLPR-based uncertainty quantification can further shed light on the accuracy of the model along the NEB pathways. In both cases, the estimated uncertainties grow quickly along the NEB path especially towards the transition state, but diminish towards the end when the system reaches the *meta* isomer products. These uncertainties are in good alignment with the actual discrepancies between the MACE-MP-0b3 predictions and the available DFT reference data, where the MACE-MP-0b3 predictions are found to notably underestimate the energies in the middle of the NEB paths, but becomes comparable when predicting the overall reaction energy.

D.4 Uncertainty Calibration

Post-processing techniques offer a way to calibrate inaccurate uncertainty estimates. A common approach involves introducing a calibration set, $\mathcal{D}_{\text{cal}} := \{(\mathbf{X}_n, \mathbf{Y}_n)\}_{n=1}^N$, to calibrate the original uncertainty. Among these techniques, conformal prediction (CP) has gained popularity for its simplicity and effectiveness in both classification and regression tasks (334–336).

To calibrate a heuristic uncertainty $\sigma(\mathbf{X})$ using CP, we first define a score function: $s(\mathbf{X}, \mathbf{Y}) := \|\tilde{\mathbf{Y}} - \mathbf{Y}\|/\sigma(\mathbf{X})$, where $\tilde{\mathbf{Y}}$ represents the model’s prediction (e.g., energy, forces, or stress). For each data point in the calibration set $\mathcal{D}_{\text{cal}}$, we compute the corresponding score $s_i = s(\mathbf{X}_i, \mathbf{Y}_i)$. Next, let $\alpha \in (0, 1)$, and define q_α as the empirical quantile of $\{s_i\}_{i=1}^N$, given by

$$q_\alpha := \text{quantile} \left(\{s_i\}_{i=1}^N, \frac{\lceil (N+1)(1-\alpha) \rceil}{N} \right).$$

The calibrated uncertainty is then obtained as $q_\alpha \sigma(\mathbf{X})$. This adjustment guarantees that the calibrated uncertainty provides a probabilistic bound on the true error, $\|\tilde{\mathbf{Y}} - \mathbf{Y}\|$, with at least $1-\alpha$ confidence (334, 337).

A recent extension of the standard CP method is to incorporate the dependence of the quantile value on the local atomic environment (LAE) (338). Specifically, the quantile forms a step function, denoted as

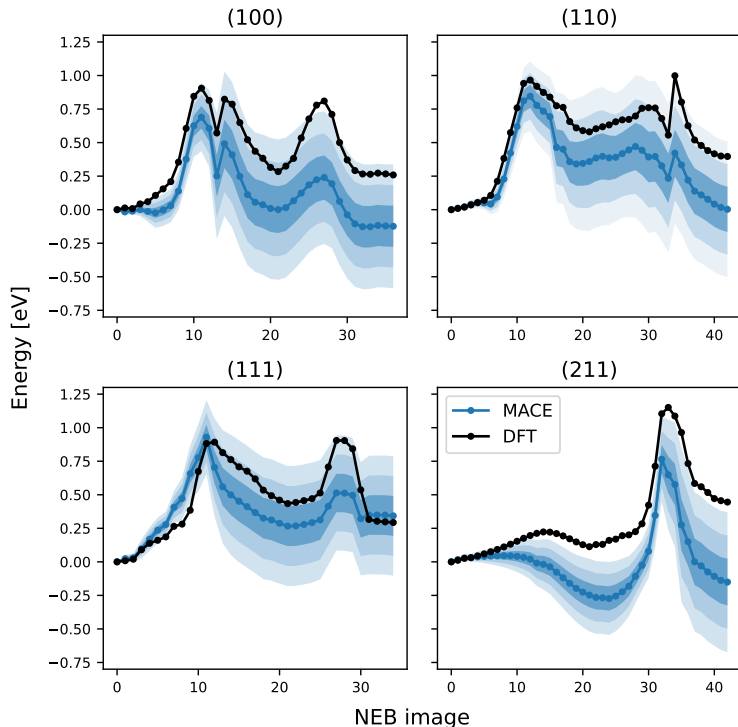


Figure S75: MACE-MP-0b3-predicted relative energies along the NEB barrier paths of CO oxidation on four different surfaces of Cu. Estimated error bounds are shaded along the plot, where 1σ , 2σ , and 3σ bounds are shown in successively lighter shades. The DFT reference data (that is not entirely consistent with the MPtrj dataset) is shown in black.

$\hat{q}_\alpha(\mathcal{G}(\mathbf{X}_i))$, where $\mathcal{G}$ is a classification model that assigns each LAE $\mathbf{X}_i$ to a distinct class. The calibration $\hat{q}_\alpha$ is then applied independently within each class, a technique referred to as class-based CP.

The hyperparameter α is a crucial component of this framework and must be carefully fined. In our numerical experiments, we set $\alpha = 0.5$ by default unless stated otherwise, as it consistently yields stable results across different scenarios. Alternatively, cross-validation on various calibration sets can be employed to determine the optimal α . The classification model $\mathcal{G}$ is constructed by clustering invariant descriptors generated by MACE-MP-0B3 using a Bayesian variant of the Gaussian Mixture Model (GMM) (339).

Next, we examine the feasibility of uncertainty calibration by applying CP and its classification-based extensions to three case studies: (1) validation error on the MP dataset, (2) NEB transition paths for CO oxidation on Cu with different surface orientations, and (3) NEB transitions in carborane rearrangement. In this section, force errors are measured using the magnitude of the force on each atom rather than considering force components, as done in the previous section.

D.4.1 Calibrating uncertainty for the MP validation set

We begin by demonstrating the improvements in predicted error using CP and class-based CP. In this example, 10% of the data was randomly selected for calibration. As shown in Figure S77, the predicted force error from LLPR (left panel) overestimates the true force error. The original CP without classification (middle panel) provides a modest improvement over LLPR. Notably, applying class-based CP (right panel) results in a further slight enhancement, primarily in terms of point density. However, increasing the number of classes in GMM clustering does not lead to any significant improvement. These findings suggest that while CP effectively improves uncertainty calibration, the additional benefit of class-based CP is marginal for large, diverse datasets. This outcome is expected, given the inherent difficulty of classifying complex atomic environments.